连续权重感知机在二分类任务中的泛化误差

LYH & WYC
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Introduction

teacher

$$y = \text{sign}\left(\frac{1}{\sqrt{n}}\mathbf{X}\mathbf{w}^\star\right)$$

$$P_{\mathbf{w}^*} \sim \mathcal{N}(0,1)$$

student

$$P(\mathbf{w}|\mathbf{y},\mathbf{X}) = \frac{P(\mathbf{y}|\mathbf{X},\mathbf{w})P(\mathbf{w})}{P(\mathbf{y},\mathbf{X})} = \frac{1}{\mathcal{Z}}P(\mathbf{y}|\mathbf{X},\mathbf{w})P(\mathbf{w})$$

$$P(\mathbf{y}|\mathbf{X},\mathbf{w}) = \mathbb{1}\left[y = \text{sign}\left(\frac{1}{\sqrt{n}}\mathbf{X}\mathbf{w}\right)\right]$$

generalization error

$$\varepsilon_{\text{gen}} = \mathbb{E}_{\mathbf{y},\mathbf{X}}\left[\Theta(-\mathbf{y}\hat{\mathbf{y}})\right] = \mathbb{E}_{\mathbf{y},\mathbf{X}}\left[\Theta(-\mathbf{z}\hat{\mathbf{z}})\right]$$

$$= \frac{1}{\pi}\arccos\left(\frac{\sigma_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sqrt{\rho_{\mathbf{w}^\star}\sigma_{\hat{\mathbf{w}}}}}\right)$$

$$= \frac{1}{\pi}\arccos\left(\sqrt{\frac{q}{\rho_{\mathbf{w}^\star}}}\right)$$

$$\rho_{\mathbf{w}^\star} = \sigma_{\mathbf{w}^*} \equiv \lim_{d\to\infty}\mathbb{E}_{\mathbf{w}^\star}\frac{1}{d}\|\mathbf{w}^\star\|_2^2$$

应该是n

Nishimori condition

$$\sigma_{\mathbf{w}^\star\hat{\mathbf{w}}} = \lim_{d\to\infty}\mathbb{E}_{\mathbf{w}^\star,\mathrm{X}}\frac{1}{d}\hat{\mathbf{w}}^\top\mathbf{w}^\star = \sigma_{\hat{\mathbf{w}}} = \lim_{d\to\infty}\mathbb{E}_{\mathbf{w}^\star,\mathrm{X}}\frac{1}{d}\|\hat{\mathbf{w}}\|_2^2$$

$$\mathbf{z} = \mathbf{X}\mathbf{w}^\star/\sqrt{n},\ \ \hat{\mathbf{z}} = \mathbf{X}\hat{\mathbf{w}}/\sqrt{n}$$

$$\sigma = \lim_{n\to\infty}\mathbb{E}_{\mathbf{w}^*,X}\frac{1}{n}\begin{bmatrix}\mathbf{w}^{*\top}\mathbf{w}^* & \mathbf{w}^{*\top}\hat{\mathbf{w}}\\ \mathbf{w}^{*\top}\hat{\mathbf{w}} & \hat{\mathbf{w}}^\top\hat{\mathbf{w}}\end{bmatrix} \equiv \begin{bmatrix}\sigma_{\mathbf{w}^*} & \sigma_{\mathbf{w}^*\hat{\mathbf{w}}}\\ \sigma_{\mathbf{w}^*\hat{\mathbf{w}}} & \sigma_{\hat{\mathbf{w}}}\end{bmatrix}$$

$$\hat{\mathbf{z}} = \sqrt{\sigma_{\hat{\mathbf{w}}}}\mathbf{x}_1$$

$$\mathbf{z} = \sqrt{\frac{\sigma_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sigma_{\hat{\mathbf{w}}}}}\mathbf{x}_1 + \sqrt{\sigma^2_{\mathrm{w}^*} - \frac{\sigma_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sigma_{\hat{\mathbf{w}}}}}\mathbf{x}_2$$

$$\varepsilon_{\text{gen}} = \mathbb{E}_{\mathbf{y},\mathbf{X}}\left[\Theta(-\mathbf{z}\hat{\mathbf{z}})\right]$$

$$= \int D\mathbf{x}_1\int D\mathbf{x}_2\Theta\left(-\sqrt{\sigma_{\hat{\mathbf{w}}}}\mathbf{x}_1\left(\sqrt{\frac{\sigma^2_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sigma_{\hat{\mathbf{w}}}}}\mathbf{x}_1 + \sqrt{\sigma^2_{\mathbf{w}^*} - \frac{\sigma^2_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sigma_{\hat{\mathbf{w}}}}}\mathbf{x}_2\right)\right)$$

$$= \int D\mathbf{x}_1\int D\mathbf{x}_2\Theta\left(-\mathbf{x}_1\right)\Theta\left(\sqrt{\frac{\sigma^2_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sigma_{\hat{\mathbf{w}}}}}\mathbf{x}_1 + \sqrt{\sigma^2_{\mathbf{w}^*} - \frac{\sigma^2_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sigma_{\hat{\mathbf{w}}}}}\mathbf{x}_2\right)$$

$$= \frac{1}{\pi}\arccos\left(\frac{\sigma_{\mathbf{w}^*\hat{\mathbf{w}}}}{\sqrt{\rho_{\mathbf{w}^\star}\sigma_{\hat{\mathbf{w}}}}}\right)$$

Replica

$$P(\mathbf{w}|\mathbf{y},\mathbf{X}) = \frac{P(\mathbf{y}|\mathbf{X},\mathbf{w})P(\mathbf{w})}{P(\mathbf{y},\mathbf{X})} = \frac{1}{\mathcal{Z}}P(\mathbf{y}|\mathbf{X},\mathbf{w})P(\mathbf{w})$$

partition function

$$\mathcal{Z}(\mathbf{y},\mathbf{X}) = P(\mathbf{y},\mathbf{X}) = \int \mathrm{d}\mathbf{w}\; P(\mathbf{w},\mathbf{y},\mathbf{X}) = \int \mathrm{d}\mathbf{w}\; P(\mathbf{y}|\mathbf{w},\mathbf{X})P(\mathbf{w})$$

$$= \int \mathrm{d}\mathbf{z}\; P(\mathbf{y}|\mathbf{z}) \int \mathrm{d}\mathbf{w}\; P(\mathbf{w})\, \delta\left(\mathbf{z} - \frac{1}{\sqrt{n}}\mathbf{w}\mathbf{X}\right)$$

$$\Phi = \frac{1}{n}\mathbb{E}_{\mathbf{y},\mathbf{X}}\; \log \mathcal{Z}(\mathbf{y},\boldsymbol{X}) \xrightarrow{\text{Replica}} \Phi = \frac{1}{n}\lim_{r\to 0}\frac{\partial \log \mathbb{E}_{\mathbf{y},\mathbf{X}}\; [\mathcal{Z}(\mathbf{y},\boldsymbol{X})^r]}{\partial r}$$

$$\mathbb{E}_{\mathbf{y},\mathrm{X}}\left[\mathcal{Z}(\mathbf{y},\mathrm{X})^r\right] = \mathbb{E}_{\mathbf{w}^*,\mathrm{X}}\left[\prod_{a=1}^{r}\int_{\mathbb{R}^n} \mathrm{d}\mathbf{z}^a P_{\mathrm{out}^a}\left(\mathbf{y}\mid\mathbf{z}^a\right)\int_{\mathbb{R}^d} \mathrm{d}\mathbf{w}^a P_{\mathrm{w}^a}\left(\mathbf{w}^a\right)\delta\left(\mathbf{z}^a - \frac{1}{\sqrt{d}}\mathrm{X}\mathbf{w}^a\right)\right]$$

$$= \mathbb{E}_{\mathrm{X}}\int_{\mathbb{R}^n} \mathrm{d}\mathbf{y} P(\mathbf{y}\,|\,X)\int_{\mathbb{R}^n} \mathrm{d}\mathbf{z}^a P_{\mathrm{out}^a}\left(\mathbf{y}\mid\mathbf{z}^a\right)\int_{\mathbb{R}^d} \mathrm{d}\mathbf{w}^a P_{\mathrm{w}^a}\left(\mathbf{w}^a\right)\delta\left(\mathbf{z}^a - \frac{1}{\sqrt{d}}\mathrm{X}\mathbf{w}^a\right)$$

$$= \mathbb{E}_{\mathrm{X}}\boxed{\int_{\mathbb{R}^n} \mathrm{d}\mathbf{y}\int_{\mathbb{R}^n} \mathrm{d}\mathbf{z}^* P_{\mathrm{out}^*}\left(\mathbf{y}\mid\mathbf{z}^*\right)\int_{\mathbb{R}^d} \mathrm{d}\mathbf{w}^* P_{\mathrm{w}^*}\left(\mathbf{w}^*\right)\delta\left(\mathbf{z}^* - \frac{1}{\sqrt{d}}\mathbf{X}^*\right)}$$

Average over y
$P(\mathbf{y}\mid X)$

$$\times\left[\prod_{a=1}^{r}\int_{\mathbb{R}^n} \mathrm{d}^a P_{\mathrm{out}^a}\left(\mathbf{y}\mid\mathbf{z}^a\right)\int_{\mathbb{R}^d} \mathrm{d}\mathbf{w}^a P_{\mathrm{w}^a}\left(\mathbf{w}^a\right)\delta\left(\mathbf{z}^a - \frac{1}{\sqrt{d}}\mathrm{X}\mathbf{w}^a\right)\right]$$

$$= \mathbb{E}_{\mathrm{X}}\int_{\mathbb{R}^n} \mathrm{d}\mathbf{y}\prod_{a=0}^{r}\int_{\mathbb{R}^n} \mathrm{d}\mathbf{z}^a P_{\mathrm{out}^a}\left(\mathbf{y}\mid\mathbf{z}^a\right)\int_{\mathbb{R}^d} \mathrm{d}\mathbf{w}^a P_{\mathrm{w}^a}\left(\mathbf{w}^a\right)\delta\left(\mathbf{z}^a - \frac{1}{\sqrt{d}}\mathrm{X}\mathbf{w}^a\right)$$

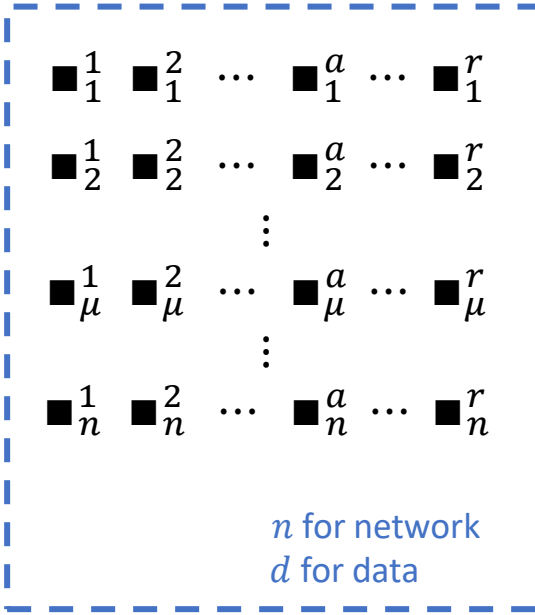
Replica

Assuming X is i.i.d, according to the central limit theorem

$$z_\mu^a = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} x_i^{(\mu)} w_i^a \sim \mathcal{N}\left(\mathbb{E}_{\mathrm{X}}\left[z_\mu^a\right], \mathbb{E}_{\mathrm{X}}\left[z_\mu^a z_\mu^b\right]\right) \qquad a,b=0..r$$

$$\mathbb{E}_{\mathrm{X}}\left[z_\mu^a\right] = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}_{\mathrm{X}}\left[x_i^{(\mu)}\right] w_i^a = 0$$

$$\mathbb{E}_{\mathrm{X}}\left[z_\mu^a z_\mu^b\right] = \frac{1}{n} \sum_{ij} \mathbb{E}_{\mathrm{X}}\left[x_i^{(\mu)} x_j^{(\mu)}\right] w_i^a w_j^b = \frac{1}{n} \sum_{ij} \delta_{ij} w_i^a w_j^b = \frac{1}{n} \mathbf{w}^a \cdot \mathbf{w}^b \equiv Q$$



The integration of column vectors → row vectors.

$$\tilde{\mathbf{z}}_\mu \equiv \left(z_\mu^a\right)_{a=0\cdots r}, \quad \tilde{\mathbf{w}}_i \equiv \left(w_i^a\right)_{a=0\cdots r}$$

$$\tilde{\mathbf{z}}_\mu \sim P_{\tilde{\mathbf{z}}}(\tilde{\mathbf{z}}; Q) = \mathcal{N}_{\tilde{\mathbf{z}}}\left(\mathbf{0}_{r+1}, Q\right) \qquad P_{\tilde{\mathrm{w}}}(\tilde{\mathbf{w}}) = \prod_{a=0}^{r} P_{\mathrm{w}}\left(\tilde{w}^a\right)$$

$$\mathbb{E}_{\mathbf{y},\mathbf{X}}\left[\mathcal{Z}(\mathbf{y},\mathbf{X})^r\right] = \mathbb{E}_{\mathbf{X}} \int \mathrm{d}\mathbf{y} \prod_{a=0}^{r} \int \mathrm{d}\mathbf{z}^a P_{\mathrm{out}^a}\left(\mathbf{y} \mid \mathbf{z}^a\right) \int \mathrm{d}\mathbf{w}^a P_{\mathbf{w}^a}\left(\mathbf{w}^a\right) \delta\left(\mathbf{z}^a - \frac{1}{\sqrt{d}} \mathbf{X}\mathbf{w}^a\right)$$

$$= \left[\int \mathrm{d}y \int \mathrm{d}\tilde{\mathbf{z}} P_{\mathrm{out}}(y \mid \tilde{\mathbf{z}}) P_{\tilde{\mathbf{z}}}(\tilde{\mathbf{z}}; Q(\tilde{\mathbf{w}}))\right]^d \left[\int \mathrm{d}\tilde{\mathbf{w}} P_{\tilde{\mathrm{w}}}(\tilde{\mathbf{w}})\right]^n$$

Replica

Introduce Fourier transform of Dirac δ function

$$1 = \int \mathrm{d}Q \prod_{a\le b} \delta\left(nQ_{ab} - \sum_{i=1}^{n} w_i^a w_i^b\right)$$

$$\propto \int \mathrm{d}Q \int \mathrm{d}\hat{Q} \prod_{a\le b} \exp\left\{-\hat{Q}_{ab}\left(nQ_{ab} - \sum_{i=1}^{n} w_i^a w_i^b\right)\right\}$$

$$\propto \int \mathrm{d}Q \int \mathrm{d}\hat{Q} \exp\left\{-\sum_{a\le b} \hat{Q}_{ab}\left(nQ_{ab} - \sum_{i=1}^{n} w_i^a w_i^b\right)\right\}$$

$$\propto \int \mathrm{d}Q \int \mathrm{d}\hat{Q} \exp(-n\,\mathrm{Tr}[Q\hat{Q}]) \exp\left(\frac{1}{2}\sum_{i=1}^{n} \tilde{\mathbf{w}}_i^\top \hat{Q} \tilde{\mathbf{w}}_i\right)$$

$$\delta(x) = \frac{1}{2i\pi}\int_{i\mathbb{R}} \mathrm{d}\hat{x} e^{-\hat{x}x}$$

$$tr(\mathbf{AB}) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij} b_{ji}$$

$a > b$:

$$\delta^2(x) = \frac{b-a}{2\pi}\delta(x), \qquad x \in [a,b]$$

Insert it into the partition function

$$\mathbb{E}_{\mathbf{y},\mathbf{X}}\left[\mathcal{Z}(\mathbf{y},\mathbf{X})^r\right] \propto \int \mathrm{d}Q \int \mathrm{d}\hat{Q}\ \exp(-n\,\mathrm{Tr}[Q\hat{Q}]) \exp\left(\frac{1}{2}\sum_{i=1}^{n} \tilde{\mathbf{w}}_i^\top \hat{Q} \tilde{\mathbf{w}}_i\right)$$

$$\left[\int \mathrm{d}y \int \mathrm{d}\tilde{\mathbf{z}}\ P_{\text{out}}(y \mid \tilde{\mathbf{z}}) P_{\tilde{\mathbf{z}}}(\tilde{\mathbf{z}}; Q(\tilde{\mathbf{w}}))\right]^d \left[\int \mathrm{d}\tilde{\mathbf{w}} P_{\tilde{\mathbf{w}}}(\tilde{\mathbf{w}})\right]^n$$

$$\propto \iint \mathrm{d}Q\, \mathrm{d}\hat{Q}\ e^{n\Phi^{(r)}(Q,\hat{Q})}$$

$$\Phi = \frac{1}{n}\lim_{r\to 0} \frac{\partial \log \mathbb{E}_{\mathbf{y},\mathbf{X}}\left[\mathcal{Z}(\mathbf{y},\mathbf{X})^r\right]}{\partial r}$$

where

$$\Phi^{(r)}(Q,\hat{Q}) = -\,\mathrm{Tr}[Q\hat{Q}] + \log \Psi_{\text{w}}^{(r)}(\hat{Q}) + \alpha \log \Psi_{\text{out}}^{(r)}(Q)$$

$$\Psi_{\text{w}}^{(r)}(\hat{Q}) = \int \mathrm{d}\tilde{\mathbf{w}}\ P_{\tilde{\mathbf{w}}}(\tilde{\mathbf{w}}) e^{\frac{1}{2}\tilde{\mathbf{w}}\hat{Q}\tilde{\mathbf{w}}}$$

$$\Psi_{\text{out}}^{(r)}(Q) = \int \mathrm{d}y \int \mathrm{d}\tilde{\mathbf{z}}\ P_{\tilde{z}}(\tilde{\mathbf{z}}; Q) P_{\text{out}}(y \mid \tilde{\mathbf{z}})$$

Laplace approximation

$$\Phi(\alpha) = \mathrm{extr}_{Q,\hat{Q}}\left\{\lim_{r\to 0}\frac{\partial \Phi^{(r)}(Q,\hat{Q})}{\partial r}\right\}$$

Replica

Replica symmetric ansatz

$$Q_{\rm rs} = \begin{pmatrix} Q^0 & m & \dots & m \\ m & Q & \dots & \dots \\ \dots & \dots & \dots & q \\ m & \dots & q & Q \end{pmatrix} \quad \text{and} \quad \hat{Q}_{\rm rs} = \begin{pmatrix} \hat{Q}^0 & \hat{m} & \dots & \hat{m} \\ \hat{m} & -\frac{1}{2}\hat{Q} & \dots & \dots \\ \dots & \dots & \dots & \hat{q} \\ \hat{m} & \dots & \hat{q} & -\frac{1}{2}\hat{Q} \end{pmatrix}$$

$$m = \frac{1}{n}\mathbf{w}^a \cdot \mathbf{w}^\star$$

$$q = \frac{1}{n}\mathbf{w}^a \cdot \mathbf{w}^b$$

$$Q = \frac{1}{n}||\mathbf{w}^a||_2^2$$

$$Q^0 = \rho_{\mathbf{w}^\star} = \frac{1}{n}||\mathbf{w}^\star||_2^2$$

$$\Phi(\alpha) = \text{extr}_{Q,\hat{Q}} \left\{ \lim_{r\to 0} \frac{\partial \Phi^{(r)}(Q,\hat{Q})}{\partial r} \right\}$$

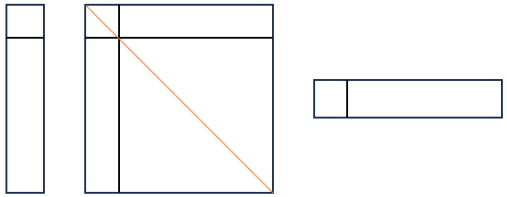
$$\Phi^{(r)}(Q,\hat{Q}) = -\operatorname{Tr}[Q\hat{Q}] + \log \Psi_{\rm w}^{(r)}(\hat{Q}) + \alpha \log \Psi_{\rm out}^{(r)}(Q)$$

- $\operatorname{Tr}\left(Q\hat{Q}\right)\Big|_{\rm rs} = Q^0\hat{Q}^0 + rm\hat{m} - \frac{1}{2}rQ\hat{Q} + \frac{r(r-1)}{2}q\hat{q}$

$$tr(\mathbf{AB}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij}b_{ji}$$

$$\begin{aligned}
&\text{extr}\left\{\lim_{r\to 0}\frac{\partial}{\partial r}\left(-\operatorname{Tr}[Q\hat{Q}]\right)\right\} \\
&= \text{extr}\left\{\lim_{r\to 0}\frac{\partial}{\partial r}\left(-Q^0\hat{Q}^0 - rm\hat{m} + \frac{1}{2}rQ\hat{Q} - \frac{r(r-1)}{2}q\hat{q}\right)\right\} \\
&= \text{extr}\left\{\lim_{r\to 0}\left(-m\hat{m} + \frac{1}{2}Q\hat{Q} - \left(r - \frac{1}{2}\right)q\hat{q}\right)\right\} \\
&= \text{extr}\left\{-m\hat{m} + \frac{1}{2}Q\hat{Q} + \frac{1}{2}q\hat{q}\right\}
\end{aligned}$$

Replica



- $\Psi_{\mathrm{w}}^{(r)}(\hat{Q}) = \int \mathrm{d}\tilde{\mathbf{w}}\; P_{\tilde{\mathbf{w}}}(\tilde{\mathbf{w}}) e^{\frac{1}{2}\tilde{\mathbf{w}}\hat{Q}\tilde{\mathbf{w}}}$

$$\boxed{\tilde{\mathbf{w}}\hat{Q}_{\mathrm{rs}}\tilde{\mathbf{w}} = w^\star \hat{Q}^0 w^\star + 2\sum_{a=1}^{r} w^\star \hat{m} w^a - \left(\hat{Q}+\hat{q}\right)\sum_{a=1}^{r}(w^a)^2 + \hat{q}\left(\sum_{a=1}^{r} w^a\right)^2}$$

$$\begin{aligned}
\Psi_{\mathrm{w}}^{(r)}(\hat{Q})\Big|_{\mathrm{rs}} &= \int \mathbf{d}\tilde{\mathbf{w}} P_{\tilde{\mathbf{w}}}(\tilde{\mathbf{w}}) e^{\frac{1}{2}\tilde{\mathbf{w}}\hat{Q}_{\mathrm{rs}}\tilde{\mathbf{w}}} \\
&= \mathbb{E}_{w^\star} e^{\frac{1}{2}\hat{Q}^0 (w^\star)^2} \int \mathbf{d}\tilde{\mathbf{w}} P_{\tilde{\mathbf{w}}}(\tilde{\mathbf{w}}) e^{w^\star \hat{m}\sum_{a=1}^{r} w^a - \frac{1}{2}(\hat{Q}+\hat{q})\sum_{a=1}^{r}(\tilde{w}^a)^2 + \frac{1}{2}\hat{q}\left(\sum_{a=1}^{r} w^a\right)^2} \\
&= \mathbb{E}_{\xi,w^\star} e^{\frac{1}{2}\hat{Q}^0 (w^\star)^2} \left[\mathbb{E}_w \exp\left(\hat{m} w^\star w - \frac{1}{2}(\hat{Q}+\hat{q}) w^2 + \hat{q}^{1/2}\xi w\right)\right]^r
\end{aligned}$$

H-S transformation

$$\mathbb{E}_\xi \exp(\sqrt{a}\xi) = e^{\frac{a}{2}}$$

$$\begin{aligned}
\Psi_{\mathrm{w}}(\hat{Q}) &= \mathrm{extr}\left\{\lim_{r\to 0}\frac{\partial}{\partial r}\left(\log \Psi_{\mathrm{w}}^{(r)}(\hat{Q})\right)\right\} \\
&= \mathrm{extr}\left\{\lim_{r\to 0}\frac{\partial}{\partial r}\mathbb{E}_{\xi,w^\star} r \log\left[\mathbb{E}_w \exp\left(\hat{m} w^\star w - \frac{1}{2}(\hat{Q}+\hat{q}) w^2 + \hat{q}^{1/2}\xi w\right)\right]\right\} \\
&= \mathrm{extr}\left\{\mathbb{E}_{\xi,w^\star} \log\left[\mathbb{E}_w \exp\left(\hat{m} w^\star w - \frac{1}{2}(\hat{Q}+\hat{q}) w^2 + \hat{q}^{1/2}\xi w\right)\right]\right\} \\
&= \mathrm{extr}\left\{\mathbb{E}_{\xi,w^\star} \exp\left(-\frac{1}{2}\hat{q}^{-1}\hat{m}^2 (w^\star)^2 + \xi \hat{q}^{-\frac{1}{2}}\hat{m} w^\star\right) \log\left[\mathbb{E}_w \exp\left(-\frac{1}{2}(\hat{Q}+\hat{q}) w^2 + \hat{q}^{1/2}\xi w\right)\right]\right\}
\end{aligned}$$

decouple the teacher and student expectations

$$\boxed{\xi \leftarrow \xi + \hat{q}^{-\frac{1}{2}}\hat{m} w^\star}$$

Replica

$$\mathbb{E}_{\xi,w^\star} \exp\left(-\frac{1}{2}\hat{q}^{-1}\hat{m}^2(w^\star)^2 + \xi\hat{q}^{-\frac{1}{2}}\hat{m}w^\star\right) \log\left[\mathbb{E}_w \exp\left(-\frac{1}{2}(\hat{Q}+\hat{q})w^2 + \hat{q}^{1/2}\xi w\right)\right]$$

define $\mathcal{Z}_{\mathrm{w}}(\gamma,\Lambda) \equiv \mathbb{E}_{w\sim P_{\mathrm{w}}}\left[e^{-\frac{1}{2}\Lambda w^2+\gamma w}\right] \xrightarrow{P_{\mathrm{w}} \sim \mathcal{N}(0,1)} \dfrac{e^{\frac{\gamma^2}{2(\Lambda+1)}}}{\sqrt{\Lambda+1}}$

Bayesian optimization $\mathcal{Z}_{\mathbf{w}} = \mathcal{Z}_{\mathbf{w}^\star}$

$$\boxed{\Psi_{\mathrm{w}}(\hat{Q},\hat{m},\hat{q}) \equiv \mathbb{E}_\xi\left[\mathcal{Z}_{\mathrm{w}^\star}\left(\hat{m}\hat{q}^{-1/2}\xi, \hat{m}\hat{q}^{-1}\hat{m}\right)\log\mathcal{Z}_{\mathrm{w}}\left(\hat{q}^{1/2}\xi, \hat{Q}+\hat{q}\right)\right]}$$

- $\Psi_{\mathrm{out}}^{(r)}(Q) = \int \mathrm{d}y \int \mathrm{d}\tilde{\mathbf{z}}\, P_{\tilde{z}}(\tilde{\mathbf{z}};Q) P_{\mathrm{out}}(y \mid \tilde{\mathbf{z}})$

$$P_{\tilde{z}}(\tilde{\mathbf{z}};Q) = \frac{e^{-\frac{1}{2}\tilde{\mathbf{z}}^\intercal Q^{-1}\tilde{\mathbf{z}}}}{\det(2\pi Q)^{1/2}}$$

$$Q_{\mathrm{rs}}^{-1} = \begin{bmatrix} Q_{00}^{-1} & Q_{01}^{-1} & Q_{01}^{-1} & Q_{01}^{-1} \\ Q_{01}^{-1} & Q_{11}^{-1} & Q_{12}^{-1} & Q_{12}^{-1} \\ Q_{01}^{-1} & Q_{12}^{-1} & Q_{11}^{-1} & Q_{12}^{-1} \\ Q_{01}^{-1} & Q_{12}^{-1} & Q_{12}^{-1} & Q_{11}^{-1} \end{bmatrix}$$

$$\boxed{\begin{aligned}
Q_{00}^{-1} &= \left(Q^0 - rm(Q+(r-1)q)^{-1}m\right)^{-1} \\
Q_{01}^{-1} &= -\left(Q^0 - rm(Q+(r-1)q)^{-1}m\right)^{-1} m(q+(r-1)q)^{-1} \\
Q_{11}^{-1} &= (Q-q)^{-1} - (Q+(r-1)q)^{-1}q(Q-q)^{-1} \\
&\quad +(Q+(r-1)q)^{-1}m\left(Q^0 - rm(Q+(r-1)q)^{-1}m\right)^{-1} m(Q+(r-1)q)^{-1} \\
Q_{12}^{-1} &= -(Q+(r-1)q)^{-1}q(Q-q)^{-1} \\
&\quad +(Q+(r-1)q)^{-1}m\left(Q - rm(Q+(r-1)q)^{-1}m\right)^{-1} m(Q+(r-1)q)^{-1}
\end{aligned}}$$

$$\det Q_{\mathrm{rs}} = (Q-q)^{r-1}(Q+(r-1)q)\left(Q^0 - rm(Q+(r-1)q)^{-1}m\right)$$

Replica

$$\Psi_{\text{out}}^{(r)}(Q)\Big|_{\text{rs}} = \int \mathrm{d}y \int \mathrm{d}\tilde{\mathbf{z}} e^{-\frac{1}{2}\tilde{\mathbf{z}}^\top Q_{\text{rs}}^{-1}\tilde{\mathbf{z}} - \frac{1}{2}\log(\det(2\pi Q_{\text{rs}}))} P_{\text{out}}(y \mid \tilde{\mathbf{z}})$$

$$= \mathbb{E}_{y,\xi} \mathrm{e}^{-\frac{1}{2}\log(\det(2\pi Q_{\text{rs}}))}$$

$$\times \int \mathrm{d}z^\star P_{\text{out}\star}(y \mid z^\star)\, e^{-\frac{1}{2}Q_{00}^{-1}(z^\star)^2} \left[\int dz P_{\text{out}}(y \mid z) e^{-Q_{01}^{-1}z^\star z - \frac{1}{2}(Q_{11}^{-1} - Q_{12}^{-1})z^2 - Q_{12}^{-1/2}\xi z}\right]^r$$

similar operation

$$\Psi_{\text{out}}(Q, m, q; \rho_{\text{w}\star}) \equiv \mathbb{E}_{y,\xi}\left[\mathcal{Z}_{\text{out}\star}\left(y, mq^{-1/2}\xi, \rho_{\text{w}\star} - mq^{-1}m\right) \log \mathcal{Z}_{\text{out}}\left(y, q^{1/2}\xi, Q - q\right)\right]$$

$$\begin{aligned}
\mathcal{Z}_{\text{out}}(y, \omega, V) &= \int dz \frac{1}{\sqrt{2\pi\Delta}} \exp\left(\frac{y - \text{sign}(z)}{2\Delta}\right) \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{(z-\omega)^2}{2V}\right) \\
&= \int_0^\infty dz \frac{1}{\sqrt{2\pi\Delta}} \exp\left(\frac{y-1}{2\Delta}\right) \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{(z-\omega)^2}{2V}\right) \\
&+ \int_{-\infty}^0 dz \frac{1}{\sqrt{2\pi\Delta}} \exp\left(\frac{y+1}{2\Delta}\right) \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{(z-\omega)^2}{2V}\right) \\
&= \mathcal{N}_y(1, \Delta^*)\left(1 + \text{erf}\left(\frac{\omega}{\sqrt{2V}}\right)\right) + \mathcal{N}_y(-1, \Delta^*)\frac{1}{2}\left(1 - \text{erf}\left(\frac{\omega}{\sqrt{2V}}\right)\right)
\end{aligned}$$

Finally, the replica symmetric solution of free energy is obtained as follows

$$\Phi_{\text{rs}}(\alpha) = \text{extr}_{Q,\hat{Q},q,\hat{q},m,\hat{m}}\left\{-m\hat{m} + \frac{1}{2}Q\hat{Q} + \frac{1}{2}q\hat{q} + \Psi_{\text{w}}(\hat{Q}, \hat{m}, \hat{q}) + \alpha\Psi_{\text{out}}(Q, m, q; \rho_{\text{w}\star})\right\}$$

Nishimori condition $Q = \rho_{\text{w}\star}, \quad m = q = q, \quad \hat{Q} = 0, \quad \hat{m} = \hat{q} = \hat{q}$

$$\Phi(\alpha) = \text{extr}_{q,\hat{q}}\left\{-\frac{1}{2}q\hat{q} + \Psi_{\text{w}}(\hat{q}) + \alpha\Psi_{\text{out}}(q; \rho_{\text{w}\star})\right\}$$

Replica

$$\hat{q} = -2\alpha\partial_q \Psi_{\text{out}}$$
$$q = -2\partial_{\hat{q}} \Psi_{\text{w}}$$

$$\Psi_{\text{w}}(\hat{q}) = \mathbb{E}_\xi \left[\mathcal{Z}_{\text{w}^\star}\left(\hat{q}^{1/2}\xi, \hat{q}\right) \log \mathcal{Z}_{\text{w}^\star}\left(\hat{q}^{1/2}\xi, \hat{q}\right) \right],$$

$$\Psi_{\text{out}}\ (q; \rho_{\text{w}^\star}) = \mathbb{E}_{y,\xi} \left[\mathcal{Z}_{\text{out}\ \star}\left(y, q^{1/2}\xi, \rho_{\text{w}^\star} - q\right) \log \mathcal{Z}_{\text{out}\ \star}\left(y, q^{1/2}\xi, \rho_{\text{w}^\star} - q\right) \right]$$

$$\mathcal{Z}_{\text{w}}(\gamma, \Lambda) \equiv \mathbb{E}_{w\sim P_{\text{w}}} \left[e^{-\frac{1}{2}\Lambda w^2 + \gamma w} \right] = \frac{e^{\frac{\gamma^2}{2(\Lambda+1)}}}{\sqrt{\Lambda+1}}$$

$$\mathcal{Z}_{\text{out}}(y, \omega, V) = \int dz \frac{1}{\sqrt{2\pi\Delta}} \exp\left(\frac{y - \text{sign}(z)}{2\Delta}\right) \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{(z-\omega)^2}{2V}\right)$$
$$= \mathcal{N}_y(1, \Delta^*)\left(1 + \text{erf}\left(\frac{\omega}{\sqrt{2V}}\right)\right) + \mathcal{N}_y(-1, \Delta^*)\frac{1}{2}\left(1 - \text{erf}\left(\frac{\omega}{\sqrt{2V}}\right)\right)$$

define

$$f_{\text{w}^\star}(\gamma, \Lambda) \equiv \partial_\gamma \log \mathcal{Z}_{\text{w}^\star}(\gamma, \Lambda) = \frac{\gamma}{1+\Lambda}$$

$$f_{\text{out}^\star}(y, \omega, V) \equiv \partial_\omega \log \mathcal{Z}_{\text{out}^\star}(y, \omega, V) = \frac{\mathcal{N}_y(1, \Delta^\star) - \mathcal{N}_y(-1, \Delta^\star)}{\mathcal{Z}_{\text{out}\ \star}(y, \omega, V)} \mathcal{N}_\omega(0, V)$$

equivalent to the definition in AMP

obtain

$$\hat{q} = \alpha \mathbb{E}_{y,\xi} \left[\mathcal{Z}_{\text{out}^\star}\left(y, q^{1/2}\xi, \rho_{\text{w}^\star} - q\right) f_{\text{out}^\star}\left(y, q^{1/2}\xi, \rho_{\text{w}^\star} - q\right)^2 \right]$$

$$q = \mathbb{E}_\xi \left[\mathcal{Z}_{\text{w}^\star}\left(\hat{q}^{1/2}\xi, \hat{q}\right) f_{\text{w}^\star}\left(\hat{q}^{1/2}\xi, \hat{q}\right)^2 \right]$$

Analytic solution

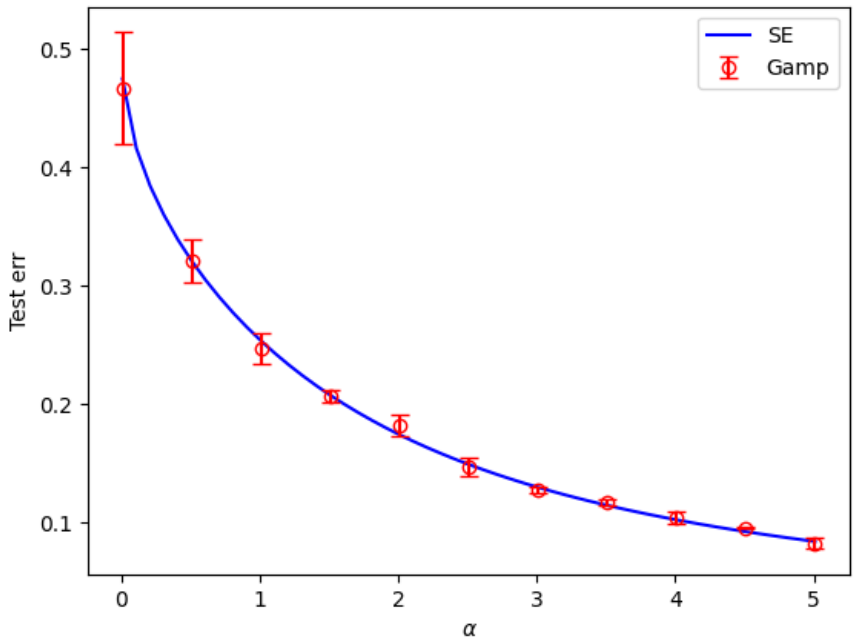
$$q = \frac{\hat{q}}{1+\hat{q}}$$

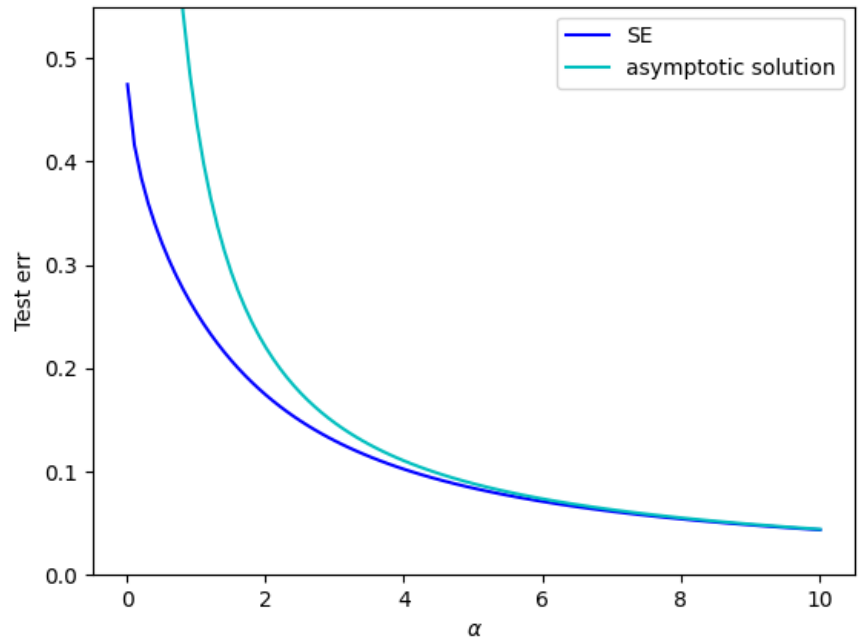
$$\hat{q} = \alpha \mathbb{E}_{y,\xi} \left[\mathcal{Z}_{\text{out}\ \star}\left(y, q^{1/2}\xi, 1-q\right) f_{\text{out}\ \star}\left(y, q^{1/2}\xi, 1-q\right)^2 \right]$$
$$= 2\alpha \int D\xi y^2 \frac{\mathcal{N}_{\sqrt{q}\xi}(0, 1-q)^2}{\frac{1}{2}\left(1 + \text{erf}\left(\frac{\sqrt{q}\xi}{\sqrt{2(1-q)}}\right)\right)}$$
$$= \frac{2}{\pi}\frac{\alpha}{1-q} \int D\xi \frac{e^{-\frac{q_{\text{b}}\xi^2}{1-q}}}{\left(1 + \text{erf}\left(\frac{\sqrt{q}\xi}{\sqrt{2(1-q)}}\right)\right)}$$

same as SE equation

Conclusion

$$e_{\mathrm{g}}^{\mathrm{bayes}}(\alpha) = \frac{1}{\pi}\,\mathrm{acos}\left(\sqrt{q_{\mathrm{b}}}\right) \underset{\alpha\to\infty}{\simeq} \frac{1}{k\pi}\frac{1}{\alpha} \simeq \frac{0.4417}{\alpha}$$





Asymptotic solution $\alpha \to \infty,\ q \to 1$

$$\int D\xi \frac{e^{-\frac{q_{\mathrm{b}}\xi^2}{1-q_{\mathrm{b}}}}}{\left(1+\mathrm{erf}\left(\frac{\sqrt{q_{\mathrm{b}}}\xi}{\sqrt{2(1-q_{\mathrm{b}})}}\right)\right)} = \int \mathrm{d}\xi \frac{\frac{-e^{\frac{\xi^2(q_{\mathrm{b}}+1)}{2(1-q_{\mathrm{b}})}}}{\sqrt{2\pi}}}{\left(1+\mathrm{erf}\left(\frac{\sqrt{q_{\mathrm{b}}}\xi}{\sqrt{2(1-q_{\mathrm{b}})}}\right)\right)} \simeq \int \mathrm{d}\xi \frac{\frac{-e^{\frac{\xi^2}{1-q_{\mathrm{b}}}}}{\sqrt{2\pi}}}{\left(1+\mathrm{erf}\left(\frac{\xi}{\sqrt{2(1-q_{\mathrm{b}})}}\right)\right)}$$

$$= \frac{\sqrt{1-q_{\mathrm{b}}}}{\sqrt{2\pi}} \int \mathrm{d}\eta \frac{e^{-\eta^2}}{1+\mathrm{erf}\left(\frac{\eta}{\sqrt{2}}\right)} = \frac{c_0}{\sqrt{2\pi}}\sqrt{1-q_{\mathrm{b}}}$$

$$q_{\mathrm{b}} = \frac{1}{2}\left(\alpha k\sqrt{\alpha^2k^2+4} - \alpha^2k^2\right) \underset{\alpha\to\infty}{\simeq} 1 - \frac{1}{\alpha^2k^2}, \quad \hat{q}_{\mathrm{b}} = k^2\alpha^2$$

$$k \equiv \frac{2c_0}{\pi\sqrt{2\pi}} \simeq 0.720647$$